

MATHEMATICS PRACTICE WORKSHEET

Topic: Height and Distance (Class 10 - Level 7/10)

Name: _____ Date: _____

Max Marks: _____ / 15

INSTRUCTIONS

Solve the following trigonometry application problems step-by-step. Show all necessary geometric diagrams and trigonometric calculations clearly.

Question 1

From the top of a 75 m high lighthouse, the angles of depression of two ships moving directly towards it are observed to be 30° and 45° . If one ship is directly behind the other on the same side of the lighthouse:

1. Find the distance between the two ships. (Take $\sqrt{3} = 1.732$)
2. If the front ship takes 10 minutes to reach the base of the lighthouse from the 45° point, find its uniform speed in m/min.

Question 2 (Q4)

A statue 1.6 m tall stands on the top of a pedestal. From a point on the ground, the angle of elevation of the top of the statue is 60° , and from the same point, the angle of elevation of the top of the pedestal is 45° .

1. Find the height of the pedestal.
2. If an observer walks 2 m closer to the pedestal from this point, find the new angle of elevation of the top of the pedestal.

Question 3 (Q5)

A hot air balloon rising vertically upwards at a constant speed of 5 m/s is observed from a point on the ground 100 m away from the launch pad. At $t = 0$, the balloon leaves the ground.

1. Find the angle of elevation of the balloon after 20 seconds.
2. How many additional seconds will it take for the angle of elevation to become 60° ? (Leave your answer in terms of $\sqrt{3}$).

KEY PROBLEM-SOLVING STRATEGY

Trigonometric Tip: Always draw the right-angled triangle representing the observer's line of sight first. Remember that *Angle of Depression* = *Angle of Elevation* due to alternate interior angles between parallel horizontal planes.